3xx Redirection HTTP Status Codes (301, 302, 304)

What are 3xx Status Codes?

Meaning: Further actions need to be taken to complete the request, usually a subsequent request to a different URL.

Category: 3xx (Redirection)

These status codes tell the client that the requested resource has moved or that a different URL should be used.

3xxx Status Codes

<u>Status Code 301:</u>

- **Name:** Moved Permanently
- **Meaning:** Resource moved permanently to a new URL. Client should use new URL for all further requests
- **When to Use:** API version changes (v1 → v2), URL structure changes, product renamed
- **ASP.NET Core Helper:** RedirectPermanent()

<u>Status Code 302:</u>

- **Name:** Found (Temporary Redirect)
- **Meaning:** Resource temporarily located at a different URL. Client should continue using original URL
- **When to Use:** Temporary maintenance, seasonal offers, A/B testing, logged-in user redirect
- **ASP.NET Core Helper:** Redirect()

<u>Status Code 304:</u>

- **Name:** Not Modified
- **Meaning:** Resource has not changed since last request. Client can use cached version
- **When to Use:** Conditional GET requests (ETag, Last-Modified headers)
- **ASP.NET Core Helper:** Not a direct helper (requires manual implementation)

Why These Status Codes Matter

Code	Client Impact	Server Responsibility
301	Client updates bookmarks/links to new URL. Search engines transfer SEO ranking to new URL	Return new URL in Location header
302	Client uses temporary URL for this request only. Keeps original URL for future requests	Return temporary URL in Location header
304	Client uses cached version, saves bandwidth. No need to download same data again	Return empty body, only headers with 304

ASP.NET Core Helper Methods for 3xx

Helper Method	Returns	Status Code
RedirectPermanent(url)	Redirects to new URL	301
RedirectPermanentPermanent (alternative)	Redirects to new URL	301
Redirect(url)	Redirects to temporary URL	302
LocalRedirect(url)	Redirects to local URL (prevents open redirect attacks)	302
RedirectToAction()	Redirects to another action	302
RedirectToRoute()	Redirects to named route	302

Notes for 304: ASP.NET Core does not have a direct helper for 304. It is handled automatically when using catching middleware with [ResponseCache] attribute or manual 'If-None-Match' header checking.

Now we are going to set up a project ‘ClothsApi2_RedirectStatusCodes’ to represent the above concepts,

Project Setup

Create the project:

Command: dotnet new webapi -n ClothsApi2_RedirectStatusCodes -f net8.0

Navigate in the project:

Command: cd ClothsApi2_RedirectStatusCodes

Install Packages

Before installing the packages, we should make sure that we are in the project folder ‘ClothsApi2_RedirectStatusCodes’ in terminal. If not, then navigate to the exact project folder in terminal.

Command 1: dotnet add package Microsoft.EntityFrameworkCore --version 8.0.0

Command 2: dotnet add package Microsoft.EntityFrameworkCore.SqlServer --version 8.0.0

Command 3: dotnet add package Microsoft.EntityFrameworkCore.Tools --version 8.0.0

Command 4: dotnet add package Microsoft.EntityFrameworkCore.Design --version 8.0.0

To check, whether the packages are successfully set up or not, we have to look over .csproj file,

ClothsApi2_RedirectStatusCodes.csproj

```
<Project Sdk="Microsoft.NET.Sdk.Web">

  <PropertyGroup>
    <TargetFramework>net8.0</TargetFramework>
    <Nullable>enable</Nullable>
    <ImplicitUsings>enable</ImplicitUsings>
  </PropertyGroup>

  <ItemGroup>
    <PackageReference Include="Microsoft.AspNetCore.OpenApi" Version="8.0.24" />
    <PackageReference Include="Microsoft.EntityFrameworkCore" Version="8.0.0" />
    <PackageReference Include="Microsoft.EntityFrameworkCore.Design" Version="8.0.0">
      <IncludeAssets>runtime; build; native; contentfiles; analyzers; buildtransitive
      </IncludeAssets>
      <PrivateAssets>all</PrivateAssets>
    </PackageReference>
    <PackageReference Include="Microsoft.EntityFrameworkCore.SqlServer" Version="8.0.0" />
    <PackageReference Include="Microsoft.EntityFrameworkCore.Tools" Version="8.0.0">
      <IncludeAssets>runtime; build; native; contentfiles; analyzers; buildtransitive
      </IncludeAssets>
      <PrivateAssets>all</PrivateAssets>
    </PackageReference>
    <PackageReference Include="Swashbuckle.AspNetCore" Version="6.6.2" />
  </ItemGroup>

</Project>
```

Step 1: Create Cloth Model

Path: ClothsApi2_RedirectStatusCodes/Models/Cloth.cs

Cloth.cs

Models > Cloth.cs > ...

```
namespace ClothsApi2_RedirectStatusCodes.Models
{
    0 references
    public class Cloth
    {
        0 references
        public int Id { get; set; }
        0 references
        public string Name { get; set; } = string.Empty;
        0 references
        public string Category { get; set; } = string.Empty;
        0 references
        public string Size { get; set; } = string.Empty;
        0 references
        public string Color { get; set; } = string.Empty;
        0 references
        public decimal Price { get; set; }
        0 references
        public bool InStock { get; set; }

        // For 304 cache validation
        0 references
        public string ETag { get; set; } = string.Empty;
    }
}
```

Step 2: Create Cloth Service (Async)

Path: ClothsApi2_RedirectStatusCodes/Services/ClothService.cs

ClothService.cs

Services > ClothService.cs > ...

```
using ClothsApi2_RedirectStatusCodes.Models;

namespace ClothsApi2_RedirectStatusCodes.Services
{
    0 references
    public static class ClothService
    {
        // In-memory data (7 cloth items)
        4 references
        private static readonly List<Cloth> _cloths = new()
        {
            new Cloth {
                Id = 1,
                Name = "Cotton T-Shirt",
                Category = "Men",
                Size = "M",
                Color = "Blue",
                Price = 599,
                InStock = true ,
                ETag = "abc123"
            },
```

```
            new Cloth {
                Id = 2,
                Name = "Denim Jacket",
                Category = "Women",
                Size = "L",
                Color = "Black",
                Price = 2499,
                InStock = true ,
                ETag = "def456"
            },
            new Cloth {
                Id = 3,
                Name = "Formal Shirt",
                Category = "Men",
                Size = "XL",
                Color = "White",
                Price = 1299,
                InStock = false,
                ETag = "ghi789"
            },
            new Cloth {
                Id = 4,
                Name = "Sports Shoes",
                Category = "Men",
                Size = "9",
                Color = "Grey",
                Price = 3499,
                InStock = true,
                ETag = "jkl012"
            },
```

```
            new Cloth {
                Id = 5,
                Name = "Silk Saree",
                Category = "Women",
                Size = "Free",
                Color = "Red",
                Price = 4999,
                InStock = true,
                ETag = "mno345"
            },
            new Cloth {
                Id = 6,
                Name = "Kids Hoodie",
                Category = "Kids",
                Size = "S",
                Color = "Yellow",
                Price = 899,
                InStock = true,
                ETag = "pqr678"
            },
            new Cloth {
                Id = 7,
                Name = "Leather Belt",
                Category = "Accessories",
                Size = "32",
                Color = "Brown",
                Price = 399,
                InStock = false,
                ETag = "stu901"
            }
        };

    // Get all cloths
    0 references
    public static async Task<List<Cloth>> GetAllClothsAsync()
    {
        await Task.Delay(100);
        return _cloths;
    }

    // Get cloth by ID
    0 references
    public static async Task<Cloth?> GetClothByIdAsync(int id)
    {
        await Task.Delay(50);
        return _cloths.FirstOrDefault(c => c.Id == id);
    }

    // Get cloth by old ID (for 301 demonstration)
    0 references
    public static async Task<Cloth?> GetClothByOldIdAsync(int oldId)
    {
        await Task.Delay(50);
        // If old ID is 100, redirect to ID 1
        if (oldId == 100)
            return _cloths.FirstOrDefault(c => c.Id == 1);
        return null;
    }
```

```
        // Get ETag for cloth (for 304 demonstration)
        0 references
        public static async Task<string?> GetClothETagAsync(int id)
        {
            await Task.Delay(30);
            var cloth = _cloths.FirstOrDefault(c => c.Id == id);
            return cloth?.ETag;
        }
    }
}
```

Step 3: Program.cs (Minimal)

Path: ClothsApi2_RedirectStatusCodes/Program.cs

Program.cs

```
var builder = WebApplication.CreateBuilder(args);

builder.Services.AddControllers();
builder.Services.AddEndpointsApiExplorer();
builder.Services.AddSwaggerGen();

var app = builder.Build();

if (app.Environment.IsDevelopment())
{
    app.UseSwagger();
    app.UseSwaggerUI();
}

app.UseHttpsRedirection();
app.UseAuthorization();
app.MapControllers();

app.Run();
```

Step 4: Controller (All 3xx Status Codes – Clean URLs)

Path: ClothsApi2_RedirectStatusCodes/Controllers/ClothController.cs

ClothController.cs

Controllers > ClothController.cs > ...

```
using ClothsApi2_RedirectStatusCodes.Models;
using ClothsApi2_RedirectStatusCodes.Services;
using Microsoft.AspNetCore.Mvc;

namespace ClothsApi2_RedirectStatusCodes.Controllers
{
    [ApiController]
    [Route("api/[controller]")]
    0 references
    public class ClothController : ControllerBase
    {
        // ==============================================================
        // 301 Moved Permanently - GET /api/Cloth/redirect-permanent/{oldId}
        // Old product URL permanently moved to new URL
        // Use RedirectPermanent() helper method
        // ==============================================================
        [HttpGet("redirect-permanent/{oldId}")]
        0 references
        public async Task<IActionResult> GetClothPermanentRedirect(int oldId)
        {
            var cloth = await ClothService.GetClothByOldIdAsync(oldId);

            if(cloth == null)
                return NotFound($"Cloth with ID {oldId} not found");

            // Returns 301 Moved Permanently with Location header
            // Search engines will transfer SEO ranking to new URL
            return RedirectPermanent($"/api/Cloth/v2/{cloth.Id}");
        }

```

```
        // ==============================================================
        // 301 Moved Permanently - GET /api/cloth/v1/{id}
        // API version changed from v1 to v2 (demonstration)
        // ==============================================================
        [HttpGet("v1/{id}")]
        0 references
        public async Task<IActionResult> GetClothV1(int id)
        {
            var cloth = await ClothService.GetClothByIdAsync(id);

            if (cloth == null)
                return NotFound($"Cloth with ID {id} not found");

            // API version permanently changed to v2
            return RedirectPermanent($"/api/Cloth/v2/{id}");
        }

```

```
45        // ==================================================================
46        // New API endpoint (v2) - Actual resource location
47        // ==================================================================
48        [HttpGet("v2/{id}")]
          0 references
49        public async Task<ActionResult<Cloth>> GetClothV2(int id)
50        {
51            var cloth = await ClothService.GetClothByIdAsync(id);
52
53            if (cloth == null)
54                return NotFound($"Cloth with ID {id} not found");
55
56            return Ok(cloth);
57        }
58
```

```
59        // ==================================================================
60        // 302 FOund (Temporary Redirect) - GET /api/Cloth/offer/summer
61        // Seasonal offers redirect to temporary URL
62        // Use Redirect() helper method
63        // ==================================================================
64        [HttpGet("offer/summer")]
          0 references
65        public IActionResult GetSummerOffer()
66        {
67            // Temporary redirect for seasonal campaign
68            // Customers should still use original URL next season
69            return Redirect("/api/Cloth/offer/summer-2026");
70        }
71
72        // ==================================================================
73        // Temporary offer page (actual content for summer 2026)
74        // ==================================================================
75        [HttpGet("offer/summer-2026")]
          0 references
76        public IActionResult GetSummerOffer2026()
77        {
78            return Ok(new
79            {
80                Offer = "Summer Sale 2026",
81                Discount = "30% off on all cotton shirts",
82                ValidUntill = "2026-18-31"
83            });
84        }
85
```

```
86      // ==============================================================
87      // 302 Found - GET /api/Cloth/category/men
88      // Redirect no filtered view (temporary URL structure)
89      // ==============================================================
90      [HttpGet("category/men")]
        0 references
91      public IActionResult GetMenCategoryRedirect()
92      {
93          // Temporary redirect while restructuring categories
94          return Redirect("/api/Cloth/filter?category=Men");
95      }
96
97      // ==============================================================
98      // Filter endpoint (actual resource location)
99      // ==============================================================
100     [HttpGet("filter")]
        0 references
101     public async Task<ActionResult<List<Cloth>>> GetClothsByFilter
102     ([FromQuery] string category)
103     {
104         var cloths = await ClothService.GetAllClothsAsync();
105         var filtered = cloths.Where(c => c.Category == category).ToList();
106
107         if (!filtered.Any())
108         {
109             return NotFound($"No cloths found in category: {category}");
110         }
111         return Ok(filtered);
112     }
113
```

```
114     // ==============================================================
115     // 304 Not Modified - GET /api/Cloth/{id}/cache
116     // Returns 304 if ETag matches (resource not changed)
117     // Client can use cached version
118     // ==============================================================
119     [HttpGet("{id}/cache")]
        0 references
120     public async Task<ActionResult<Cloth>> GetClothWithCache(int id)
121     {
122         var cloth = await ClothService.GetClothByIdAsync(id);
123
124         if (cloth == null)
125             return NotFound($"Cloth with ID {id} not found");
126
127         // Get Etag from service (simulates database version)
128         var currentETag = await ClothService.GetClothETagAsync(id);
129
130         // Check If-None-Match header from client
131         var clientETag = Request.Headers["If-None-Match"].ToString();
132
133         if (!string.IsNullOrEmpty(clientETag) && clientETag == currentETag)
134         {
135             // Resource not changed - return 304 Not Modified
136             // No response body, client uses cached version
137             return StatusCode(304);
138         }
139
140         // Set: ETag in response header for future requsts
141         Response.Headers.Append("ETag", currentETag);
142
143         return Ok(cloth);
144     }
145
```

```
146         // ==============================================================
147         // LocalRedirect - GET /api/cloth/local-redirect
148         // Safer redirect (prevents open redirect attacks)
149         // ==============================================================
150         [HttpGet("local-redirect")]
            0 references
151         public IActionResult LocalRedirectExample()
152         {
153             // Only allows local URLs (prevents redirect to malicious external sites)
154             return LocalRedirect("/api/Cloth");
155         }
156
157         // ==============================================================
158         // RedirectToAction - GET /api/Cloth/redirect-to-action
159         // Redirect to another action within same controller
160         // ==============================================================
161         [HttpGet("redirect-to-action")]
            0 references
162         public IActionResult RedirectToGetAllCloths()
163         {
164             // Redirects to GetAllCloths action (GET /api/cloth)
165             return RedirectToAction(nameof(GetAllCloths));
166         }
167
168         // ==============================================================
169         // Helper action for RedirectToAction demonstration
170         // ==============================================================
171         [HttpGet]
            1 reference
172         public async Task<ActionResult<List<Cloth>>> GetAllCloths()
173         {
174             var cloths = await ClothService.GetAllClothsAsync();
175             return Ok(cloths);
176         }
177     }
178 }
```

IMPORTANT NOTE ABOUT REDIRECT TESTING

Why we don't see 301/302 directly in browser:

When we visit a URL that returns 301 or 302, the browser **automatically follows** the redirect. We will only see the **final 200 OK** response in the browser window.

N.B.:

To see the actual 301/302 status code, we will use here the **Postman console**. **Postman** is also an API testing tool, and inside the **Postman**, there is a console to see the logs. We can see our redirect status logs here and this is the easiest way to understand the above.

We can open **Postman Console** by, Shortcut: ***Ctrl + Alt + C*** or click the "Console" icon at the bottom left.

Test 1: 301 – Permanent Redirect (Old ID 100 → v2 ID 1)

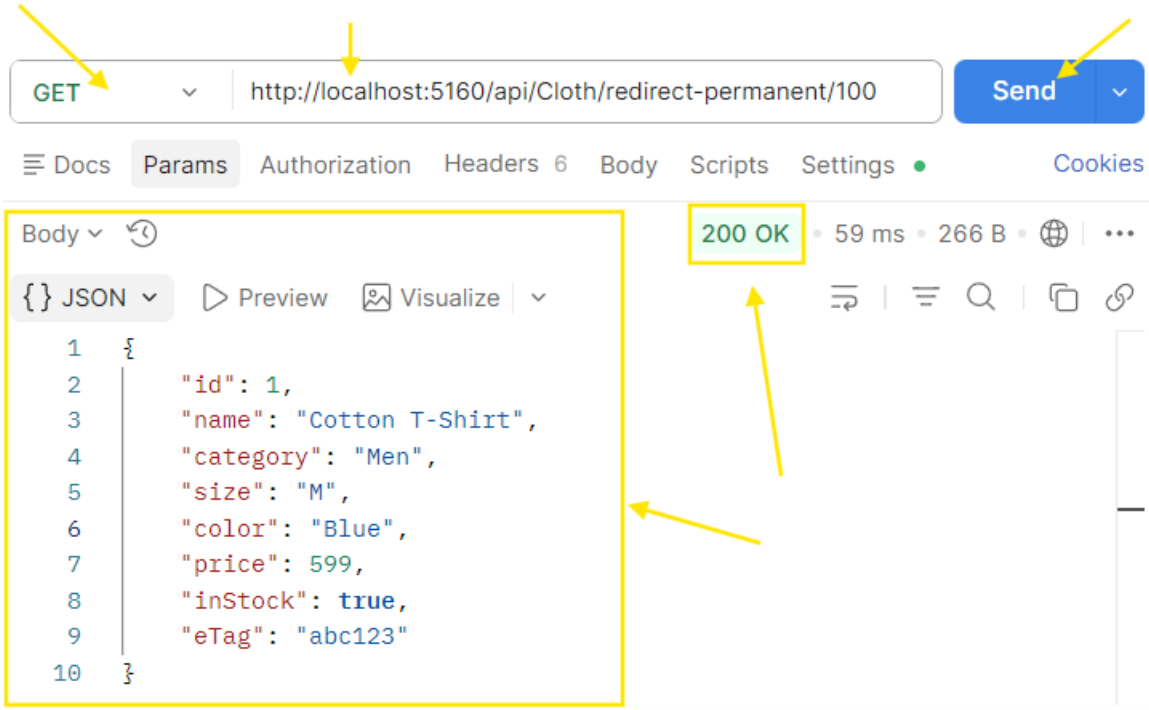
Request: GET http://localhost:5160/api/Cloth/redirect-permanent/100

Intermediate Response (What server returns first):

➔ **Postman Console output:**

Final Response (After browser auto-redirects):

➔ **Postman Response:**



Test 2: 301 – API Version Redirect (v1 ID 3 → v2 ID 3)

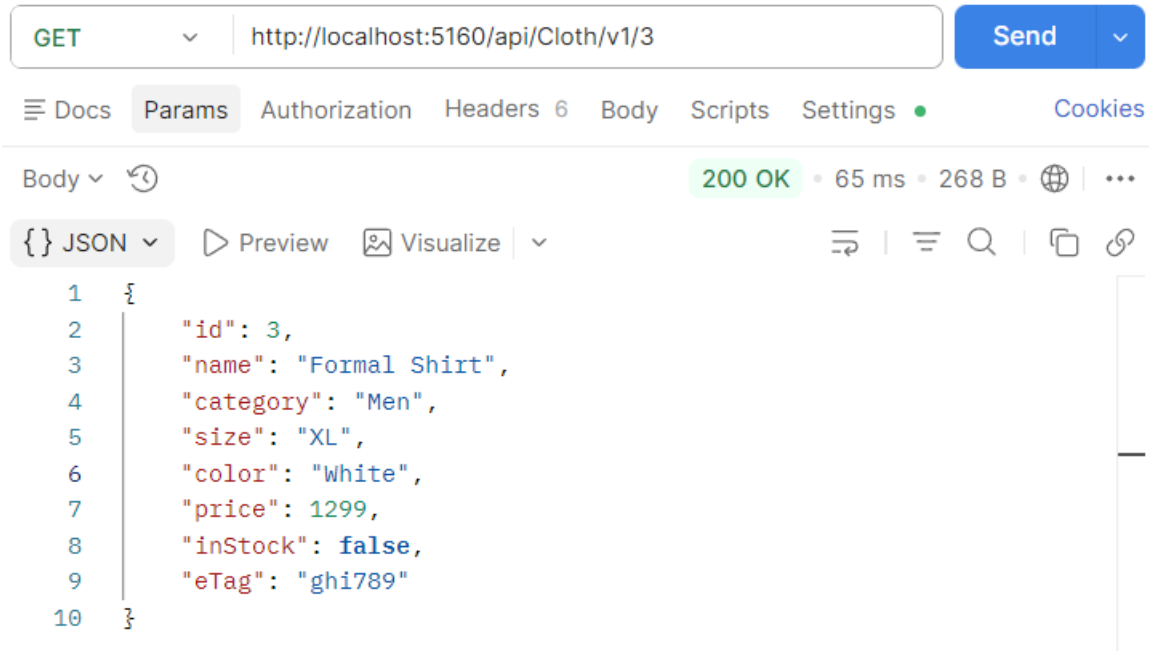
Request: GET http://localhost:5160/api/Cloth/v1/3

Intermediate Response (What server returns first):

➔ Postman Console output:

Final Response (After browser auto-redirects):

➔ Postman Response:



Test 3: 301 – Seasonal Offer Redirect (summer → summer-2026)

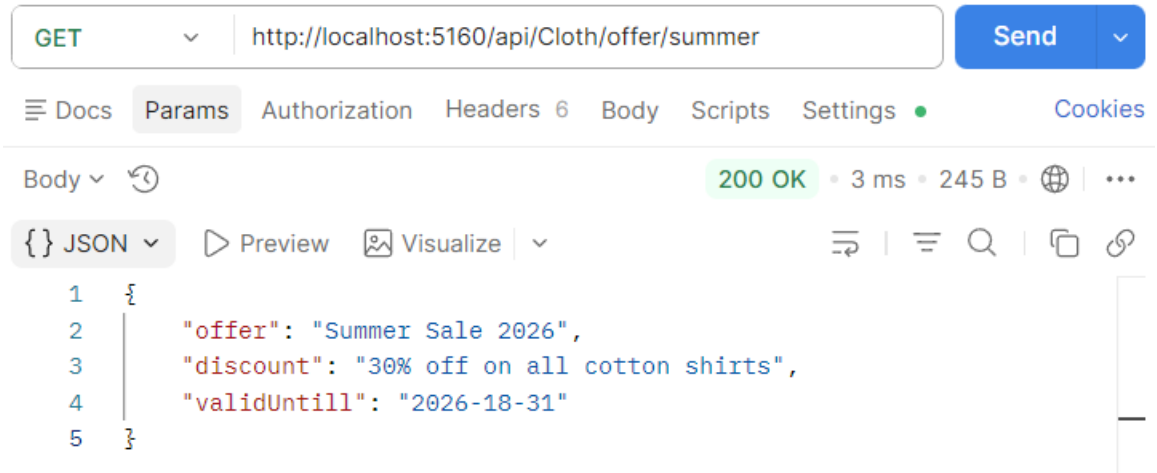
Request: GET http://localhost:5160/api/Cloth/offer/summer

Intermediate Response (What server returns first):

➔ **Postman Console output:**

Final Response (After browser auto-redirects):

➔ **Postman Response:**



Test 4: 302 – Category Redirect (category/men → filter?category=Men)

Request: GET http://localhost:5160/api/Cloth/category/men

Intermediate Response (What server returns first):

➔ Postman Console output:

```
Console   Terminal                                   All Logs   Clear
▾ GET http://localhost:5160/api/Cloth/category/men                    302
  ▸ Network                                                           Show raw log
  ▸ Request Headers
  ▾ Response Headers
    Content-Length: "0"
    Date: "Mon, 27 Apr 2026 17:18:55 GMT"
    Server: "Kestrel"
    Location: "/api/Cloth/filter?category=Men"
▸ GET http://localhost:5160/api/Cloth/filter?category=Men             200  111 ms
```

Final Response (After browser auto-redirects):

➔ Postman Response:

Body 200 OK • 128 ms • 507 B

{ } JSON Preview Visualize


```
1  [
2      {
3          "id": 1,
4          "name": "Cotton T-Shirt",
5          "category": "Men",
6          "size": "M",
7          "color": "Blue",
8          "price": 599,
9          "inStock": true,
10         "eTag": "abc123"
11     },
12     {
13         "id": 3,
14         "name": "Formal Shirt",
15         "category": "Men",
16         "size": "XL",
17         "color": "White",
18         "price": 1299,
19         "inStock": false,
20         "eTag": "ghi789"
21     },
22     {
23         "id": 4,
24         "name": "Sports Shoes",
25         "category": "Men",
26         "size": "9",
27         "color": "Grey",
28         "price": 3499,
29         "inStock": true,
30         "eTag": "jkl012"
31     }
32 ]
```

Test 5: 302 – Local Redirect (secure redirect)

Request: GET http://localhost:5160/api/Cloth/local-redirect

Intermediate Response (What server returns first):

➔ **Postman Console output:**

```
Console   Terminal                                   All Logs   Clear
▾ GET http://localhost:5160/api/Cloth/local-redirect                    302
  ▸ Network                                                             Show raw log
  ▸ Request Headers
  ▾ Response Headers
    Content-Length: "0"
    Date: "Mon, 27 Apr 2026 17:32:09 GMT"
    Server: "Kestrel"
    Location: "/api/Cloth"
▸ GET http://localhost:5160/api/Cloth                                   200  122 ms
```

Final Response (After browser auto-redirects):

➔ **Postman Response:**

Body

{} JSON Preview Visualize


```
1  [
2      {
3          "id": 1,
4          "name": "Cotton T-Shirt",
5          "category": "Men",
6          "size": "M",
7          "color": "Blue",
8          "price": 599,
9          "inStock": true,
10         "eTag": "abc123"
11     },
12     {
13         "id": 2,
14         "name": "Denim Jacket",
15         "category": "Women",
16         "size": "L",
17         "color": "Black",
18         "price": 2499,
19         "inStock": true,
20         "eTag": "def456"
21     },
22     {
23         "id": 3,
24         "name": "Formal Shirt",
25         "category": "Men",
26         "size": "XL",
27         "color": "White",
28         "price": 1299,
29         "inStock": false,
30         "eTag": "ghi789"
31     },
32     {
33         "id": 4,
34         "name": "Sports Shoes",
35         "category": "Men",
36         "size": "9",
37         "color": "Grey",
38         "price": 3499,
39         "inStock": true,
40         "eTag": "jkl012"
41     },
```

```
42      {
43          "id": 5,
44          "name": "Silk Saree",
45          "category": "Women",
46          "size": "Free",
47          "color": "Red",
48          "price": 4999,
49          "inStock": true,
50          "eTag": "mno345"
51      },
52      {
53          "id": 6,
54          "name": "Kids Hoodie",
55          "category": "Kids",
56          "size": "S",
57          "color": "Yellow",
58          "price": 899,
59          "inStock": true,
60          "eTag": "pqr678"
61      },
62      {
63          "id": 7,
64          "name": "Leather Belt",
65          "category": "Accessories",
66          "size": "32",
67          "color": "Brown",
68          "price": 399,
69          "inStock": false,
70          "eTag": "stu901"
71      }
72  ]
```

<u>Test 6: 302 – Redirect to Action (go-to-action → GetAllCloths)</u>

<u>Request:</u> GET http://localhost:5160/api/Cloth/redirect-to-action

<u>Intermediate Response (What server returns first):</u>

➔ **Postman Console output:**

Final Response (After browser auto-redirects):

➔ Postman Response:

Body

{} JSON Preview Visualize


```
[
    {
        "id": 1,
        "name": "Cotton T-Shirt",
        "category": "Men",
        "size": "M",
        "color": "Blue",
        "price": 599,
        "inStock": true,
        "eTag": "abc123"
    },
    {
        "id": 2,
        "name": "Denim Jacket",
        "category": "Women",
        "size": "L",
        "color": "Black",
        "price": 2499,
        "inStock": true,
        "eTag": "def456"
    },
    {
        "id": 3,
        "name": "Formal Shirt",
        "category": "Men",
        "size": "XL",
        "color": "White",
        "price": 1299,
        "inStock": false,
        "eTag": "ghi789"
    },
    {
        "id": 4,
        "name": "Sports Shoes",
        "category": "Men",
        "size": "9",
        "color": "Grey",
        "price": 3499,
        "inStock": true,
        "eTag": "jkl012"
    },
    {
        "id": 5,
        "name": "Silk Saree",
        "category": "Women",
        "size": "Free",
        "color": "Red",
        "price": 4999,
        "inStock": true,
        "eTag": "mno345"
    },
    {
        "id": 6,
        "name": "Kids Hoodie",
        "category": "Kids",
        "size": "S",
        "color": "Yellow",
        "price": 899,
        "inStock": true,
        "eTag": "pqr678"
    },
    {
        "id": 7,
        "name": "Leather Belt",
        "category": "Accessories",
        "size": "32",
        "color": "Brown",
        "price": 399,
        "inStock": false,
        "eTag": "stu901"
    }
]
```